MOBILITY AND TRAFFIC MANAGEMENT

During the Simhasth, Ujjain's road network will experience extraordinary and highly variable demand due to the simultaneous movement of pilgrims, residents, public transport, private vehicles, buses, emergency vehicles, service vehicles and pedestrian groups. Traffic demand will not be uniform. It will fluctuate significantly based on bathing schedules, religious events, peak arrival/departure periods, location of major venues, weather conditions, time of day and crowd density. Large pedestrian flows may also interact with vehicular traffic at multiple locations.

The temporary traffic arrangements required for the Mela further complicates the problem. Diversions, one-way systems, parking areas, shuttle services, pedestrian corridors, restricted zones and emergency routes may need to be dynamically modified based on ground conditions. Authorities therefore require solutions that can provide real-time visibility, prediction and coordinated control of mobility across the Mela area.

Some of the key challenges include

- High mobility demand from different entry points into the city (airports, railway stations, bus stands, etc.)
- Spike in vehicular volumes, change in traffic patterns and rapid breach of threshold levels
- High pedestrian-vehicle interaction and conflict.
- High parking demand and spillover into on to public roads
- Congestion caused by public service vehicles like waste collection vehicles.
- Difficulty in identification of emergency evacuation corridors and ensuring they remain accessible

Participants are expected to exhibit capabilities in technology, urban planning and/or transport planning. Submissions should propose solutions that support authorities in planning and managing seamless movement of people and vehicles and address volatile mobility demand. Proposals should include at least one of more of the following capabilities.

- Real-time congestion forecasting, detection and prediction
- Intelligent route and diversion management
- Smart parking allocation and occupancy prediction
- Public transport and shuttle optimisation
- Emergency-route identification and prioritization
- Digital twin models for scenario planning
- Offline/low connectivity operational capabilities.

These is the general direction in which applicants are encouraged to think. However, applicants are welcome to submit proposals rooted in principles that do not feature in this list but are capable of address the mobility and traffic challenges anticipated during the Simhasth.